The documentation and process conversion measures necessary to comply with this revision shall be completed by 25 December 1997

INCH POUND

MIL-PRF-19500/379E
25 September 1997
SUPERSEDING
MIL-S-19500/379D
15 December 1992

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, HIGH-POWER
TYPES 2N3791, 2N3792 JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Depart
ments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for PNP silicon, high-power transistors. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 TO-3.

1.3 Maximum ratings.

| | $P_T$ 1/ $T_A$ = +25°C | $P_T$ 2/ $T_C$ = +100°C | $V_{CBO}$ | $V_{CEO}$ | $V_{EBO}$ | $I_B$ | $I_C$ | $T_J$ and $T_{STG}$ | $R_{\theta JC}$ |
|---|---|---|---|---|---|---|---|---|---|
| | W | W | V dc | V dc | V dc | A dc | A dc | °C | °C/W |
| 2N3791 | 5.0 | 85.7 | 60 | 60 | 7.0 | 4.0 | 10 | -65 to +200 | 1.17 |
| 2N3792 | 5.0 | 85.7 | 80 | 80 | 7.0 | 4.0 | 10 | -65 to +200 | 1.17 |

1/ Derate linearly 28.57 mW/°C above $T_A$ = +25°C.

2/ Derate linearly 0.857 W/°C above $T_C$ = +100°C.

1.4 Primary electrical characteristics at $T_C$ = +25°C ± 3°C.

| | $h_{FE2}$ 1/ | $h_{FE4}$ 1/ | $V_{BE(SAT)1}$ 1/ | $V_{CE(SAT)1}$ 1/ | $C_{obo}$ | $\|h_{fe}\|$ |
|---|---|---|---|---|---|---|
| | $V_{CE}$ = 2.0 V dc<br>$I_C$ = 3.0 A dc | $V_{CE}$ = 4.0 V dc<br>$I_C$ = 10 A dc | $I_C$ = 5.0 A dc<br>$I_B$ = 0.5 A dc | $I_C$ = 5.0 A dc<br>$I_B$ = 0.5 A dc | $V_{CB}$ = 10 V dc<br>$I_E$ = 0<br>f = 1 MHz | $V_{CE}$ = 10 V dc<br>$I_C$ = 0.5 A dc<br>f = 1 MHz |
| | | | V dc | V dc | pF | |
| Min | 30 | 5.0 | | | | 4.0 |
| Max | 120 | | 1.5 | 1.0 | 500 | 20 |

1/ Pulse (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specifiecation sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.4).

3.2 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requiements and physical dimensions. The interface requiements and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 herein.

3.4.1 Lead finish. Unless otherwise specified, lead finish shall be solderable in accordance with MIL-STD-750, MIL-PRF-19500, and herein.

3.5 Marking. Devices shall be marked as specified in MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

4. VERIFICATION

4.1 Classification of and inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.



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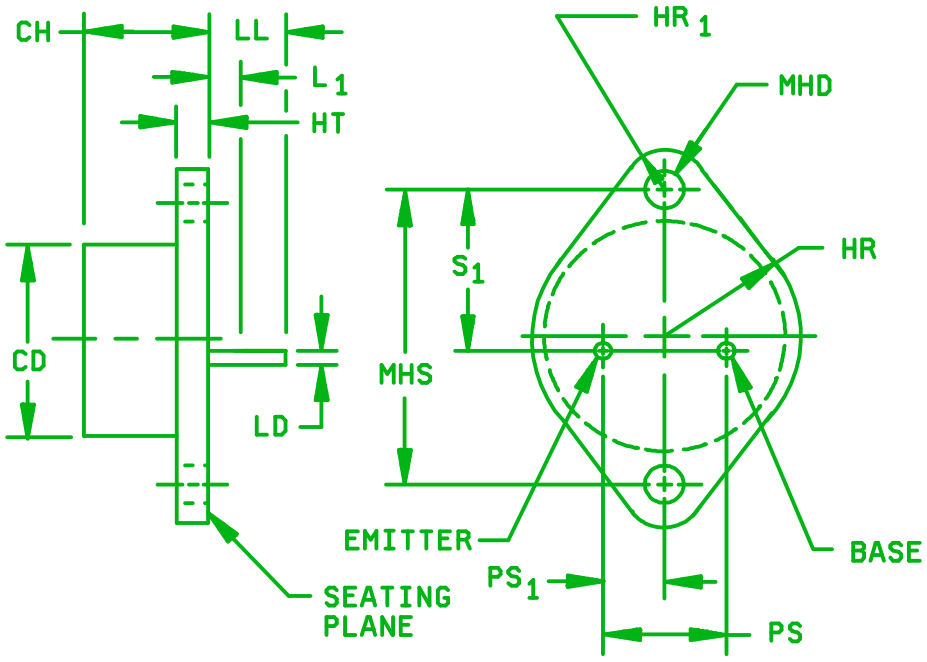



FIGURE 1. Physical dimensions TO-3.



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| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| CD | - - - | .875 | - - - | 22/22 | 6 |
| CH | .270 | .380 | 6.86 | 9.65 | |
| HT | .060 | .135 | 1.52 | 3.43 | |
| HR | .495 | .525 | 12.57 | 13.34 | |
| $HR_1$ | .131 | .188 | 3.33 | 4.78 | |
| LD | .038 | .043 | 0.97 | 1.09 | |
| LL | .312 | .500 | 7.92 | 12.70 | |
| $L_1$ | - - - | .050 | - - - | 1.27 | |
| MHD | .151 | .161 | 3.84 | 4.09 | |
| MHS | 1.177 | 1.197 | 29.90 | 30.40 | |
| PS | .420 | .440 | 10.67 | 11.18 | 3, 4 |
| $PS_1$ | .205 | .225 | 5.21 | 5.72 | 3, 4 |
| $S_1$ | .655 | .675 | 16.64 | 17.15 | 3 |

NOTES:
1. Dimensions are in inches.
2. Terminal 1, base; terminal 2, emitter; case, collector.
3. These dimensions should be measured at points .050 inch (1.27 mm) to .055 inch (1.40 mm) below seating plane. When gauge is not used, measurement will be made at the seating plane.
4. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
5. Collector shall be electrically connected to the case.
6. LD applies between $L_1$ and LL. Diameter is uncontrolled in $L_1$.

FIGURE 1. Physical dimensions. Continued.



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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (appendix E, table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table IV of MIL-PRF-19500) | Measurement | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| 9 | $I_{CES1}$ and $h_{FE2}$ | $I_{CES1}$ |
| 11 | $I_{CES1}$ and $h_{FE2}$<br>$\Delta I_{CES1}$ = 100 percent of initial value or 50 µA dc, whichever is greater.<br>$\Delta h_{FE2}$ = ± 15 percent of initial value | $I_{CES1}$ and $h_{FE2}$;<br>$\Delta I_{CES1}$ = 100 percent of initial value or 50 µA dc, whichever is greater. |
| 12 | See 4.5.31 | See 4.3.1 |
| 13 | $\Delta I_{CES1}$ = 100 percent of initial value or 1 µA dc, whichever is greater;<br>$\Delta h_{FE2}$ = ± 15 percent of initial value subgroups 2 and 3 of table I herein. | $\Delta I_{CES1}$ = 100 percent of initial value or 1 mA dc, whichever is greater;<br>$\Delta h_{FE2}$ = ± 15 percent of initial value; subgroup 2 of table I herein. |

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

$T_J$ = +187.5°C ±12.5°C; $V_{CB}$ = 35 ±5 V dc; $T_A \leq$ +100°C.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V, MIL-PRF-19500 and herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VIa (JANS) and table VIb (JANTX and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (endpoints) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, appendix E, table VIa of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| B3 | 2037 | Test condition A; All internal wires for each device shall be pulled separately. |
| B4 | 1037 | $V_{CB}$ = 30 V dc; $P_T$ = 5 W at $T_A$ = +25°C ±3°C, $t_{on}$ = $t_{off}$ = 3 minutes minimum for 2,000 cycles. No heat sink nor forced air on the device shall be permitted. |
| B5 | 1027 | $V_{CB}$ = 30 V dc; $T_A$ = +125°C ±25°C for 96 hours; $P_T$ = 5 W at $T_A$ = +112°C or adjusted as required by the chosen $T_A$ to give an average lot $T_J$ = +275°C. Marking legibility requirements shall not apply. |
| B6 | 3131 | See 4.5.2 |



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4.4.2.2 Group B inspection, appendix E, table VIb of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| B3 | 1027 | $T_J$ = +187.5°C ±12.5°C; $V_{CB}$ = 35 V dc ±5 V dc; $T_A \leq$ +100°C. |
| B5 | 3131 | See 4.5.2. |
| B6 | 1032 | $T_A$ = +200°C. |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500, and as follows. Electrical measurements (endpoints) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1 Group C inspection, table V of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| C2 | 2036 | Test condition A; weight = 10 pounds, t = 15 s. |
| C6 | 1026 | $T_C$ = +187.5°C ±12.5°C; $V_{CE}$ = 25 V dc ±5 V dc; $T_A \leq$ +100°C. |

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with test method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power application shall be 1.0 A dc minimum.

b. Collector to emitter voltage magnitude shall be 10 V dc minimum.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be +25°C ≤ $T_R$ ≤ +75°C and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to header.

f. Maximum limit of $R_{\theta JC}$ shall be 1.17°C/W.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Collector-emitter breakdown voltage | 3011 | Bias conditions D, $I_C$ = 10 mA dc; pulsed (see 4.5.1) | $V_{(BR)CEO}$ | | | |
| 2N3791 | | | | 60 | | V dc |
| 2N3792 | | | | 80 | | V dc |
| Emitter-base cutoff current | 3061 | Bias condition D; $V_{EB}$ = 7 V dc | $I_{EBO}$ | | 5.0 | mA dc |
| Collector-base cutoff current | 3036 | Bias conditions D | $I_{CBO}$ | | | |
| 2N3791 | | $V_{CB}$ = 60 V dc | | - - - | 5.0 | mA dc |
| 2N2792 | | $V_{CB}$ = 80 V dc | | - - - | 5.0 | mA dc |
| Collector-emitter cutoff current | 3041 | Bias condition A; $V_{BE}$ = 1.5 V dc | $I_{CEX}$ | | | |
| 2N3791 | | $V_{CE}$ = 60 V dc | | | 5.0 | mA dc |
| 2N3792 | | $V_{CE}$ = 80 V dc | | | 5.0 | mA dc |
| Collector-emitter cutoff current | 3041 | Bias condition C | $I_{CES1}$ | | | |
| 2N3791 | | $V_{CE}$ = 50 V dc | | | 5.0 | mA dc |
| 2N3792 | | $V_{CE}$ = 70 V dc | | | 5.0 | mA dc |
| Base-emitter saturated voltage | 3066 | Test condition A; $I_C$ = 5 A dc $I_B$ = 0.5 A dc pulsed (see 4.5.1) | $V_{BE(sat)1}$ | | 1.5 | V dc |
| Base-emitter saturated voltage | 3066 | Test condition A; $I_C$ = 10 A dc $I_B$ = 2 A dc pulsed (see 4.5.1) | $V_{BE(sat)2}$ | | 3.0 | V dc |
| Collector-emitter saturated voltage | 3071 | $I_C$ = 5 A dc; $I_B$ = 0.5 A dc; pulsed (see 4.5.1) | $V_{CE(sat)1}$ | - - - | 1.0 | V dc |
| Collector-emitter saturated voltage | 3071 | $I_C$ = 10 A dc; $I_B$ = 2 A dc; pulsed (see 4.5.1) | $V_{CE(sat)2}$ | | 2.5 | V dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 2.0 V dc; $I_C$ = 1.0 A dc; pulsed (see 4.5.1) | $h_{FE1}$ | 50 | 150 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 2.0 V dc; $I_C$ = 3.0 A dc; pulsed (see 4.5.1) | $h_{FE2}$ | 30 | 120 | |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued | | | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 2.0 V dc; $I_C$ = 5 A dc; pulsed (see 4.5.1) | $h_{FE3}$ | 10 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 4.0 V dc; $I_C$ = 10 A dc pulsed (see 4.5.1) | $h_{FE4}$ | 5 | | |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_A$ = +150°C | | | | |
| Collector to emitter cutoff current | 3041 | Bias conditions C | $I_{CES2}$ | | | |
| 2N3791 | | $V_{CE}$ = 50 V dc | | | 5.0 | mA dc |
| 2N3792 | | $V_{CE}$ = 70 V dc | | | 5.0 | mA dc |
| Low temperature operation: | | $T_A$ = -55°C | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 2.0 V dc; $I_C$ = 3.0 A dc; pulsed (see 4.5.1) | $h_{FE5}$ | 12 | | |
| Subgroup 4 | | | | | | |
| Switching parameters | | | | | | |
| Pulse delay time | | See figure 2 | $t_d$ | | 0.2 | µs |
| Pulse rise time | | See figure 2 | $t_r$ | | 1.3 | µs |
| Pulse storage time | | See figure 2 | $t_s$ | | 1.4 | µs |
| Pulse fall time | | See figure 2 | $t_f$ | | 1.0 | µs |
| $t_{off}$ | | $t_s + t_f$ | $t_{off}$ | | 2.0 | µs |
| Small-signal short-circuit forward-current transfer ratio | 3206 | $V_{CE}$ = 10 V dc; $I_C$ = 0.5 A dc f = 1 kHz | $h_{fe}$ | 30 | 300 | |
| Magnitude of small-signal short-circuit, forward-current transfer ratio | 3306 | $V_{CE}$ = 10 V dc; $I_C$ = 0.5 A dc; f = 1 MHz | $\|h_{fe}\|$ | 4.0 | 20 | - - - |
| Open circuit output capacitance | 3236 | $V_{CB}$ = 10 V dc; $I_E$ = 0; f = 1 MHz | $C_{obo}$ | | 500 | pF |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 5 | | | | | | |
| Safe operating area (continuous dc) | 3051 | $T_C$ = +25°C; t ≥ 1 s; 1 cycle; (see figure 3) | | | | |
| Test 1 | | $V_{CE}$ = 15 V dc; $I_C$ = 10 A dc | | | | |
| Test 2 | | $V_{CE}$ = 40 V dc; $I_C$ = 3.75 A dc | | | | |
| Test 3 | | | | | | |
| 2N3791 | | $V_{CE}$ = 55 V dc; $I_C$ = 0.9 A dc | | | | |
| 2N3792 | | $V_{CE}$ = 65 V dc; $I_C$ = 0.9 A dc | | | | |
| Safe operating area (clamped inductive) | 3053 | $T_A$ = +25°C; $I_C$ = 10 A dc; $V_{CC}$ = 15 V dc; (see figures 5 and 6) | | | | |
| 2N3791 | | Clamp voltage = 60 V dc | | | | |
| 2N3792 | | Clamp voltage = 80 V dc | | | | |
| Subgroups 6 and 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-PRF-19500.



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TABLE II. Groups A, B, and C electrical measurements. 1/ 2/ 3/

| Step | Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1. | Collector-emitter cutoff current<br>2N3791<br>2N3792 | 3041 | Bias condition C<br>$V_{CE}$ = 50 V dc<br>$V_{CE}$ = 70 V dc | $I_{CES1}$ | | <br>5<br>5 | <br>mA dc<br>mA dc |
| 2. | Collector-emitter (voltage saturated) | 3071 | $I_C$ = 5 A dc;<br>$I_B$ = 0.5 A dc;<br>pulsed (see 4.5.1) | $V_{CE(sat)1}$ | | 1.0 | V dc |
| 3. | Base-emitter (voltage saturated) | 3066 | Bias condition A;<br>$I_C$ = 5.0 A dc<br>$I_B$ = 0.5 A dc<br>pulsed (see 4.5.1) | $V_{BE(sat)1}$ | | 1.5 | V dc |
| 4. | Forward-current transfer ratio | 3076 | $V_{CE}$ = 2.0 V dc;<br>$I_C$ = 3.0 A dc;<br>pulsed (see 4.5.1) | $h_{FE2}$ | 30 | 120 | |
| 5. | Collector-emitter cutoff current<br>2N3791<br>2N3792 | 3041 | Bias condition C<br>$V_{CE}$ = 50 V dc<br>$V_{CE}$ = 70 V dc | $\Delta I_{CES1}$ 4/ | 100 percent of initial value or 1 µA dc; whichever is greater. | | |
| 6. | Forward-current transfer ratio | 3076 | $V_{CE}$ = 2.0 V dc;<br>$I_C$ = 3.0 A dc;<br>pulsed (see 4.5.1) | $\Delta h_{FE2}$ 4/ | ±25 percent change from initial value | | |
| 7. | Collector-emitter (voltage saturated) | 3071 | $I_C$ = 5 A dc;<br>$I_B$ = 0.5 A dc;<br>pulsed (see 4.5.1) | $\Delta V_{CE(sat)1}$ 4/ | ±50 mV dc change from previously measured value | | |

1/ The electrical measurements for appendix E, table VIa (JANS) of MIL-PRF-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1, 2, 3, and 4.
b. Subgroup 4, see table II herein, steps 1, 2, 3, 4, and 7.
c. Subgroup 5, see table II herein, steps 1, 2, 3, 4, 5, 6, and 7.

2/ The electrical measurements for appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table II, steps 1 and 4.
b. Subgroup 3, see table II herein, steps 1, 4, 5, and 6.
c. Subgroup 6, see table II herein, steps 1, 4, 5, and 6.

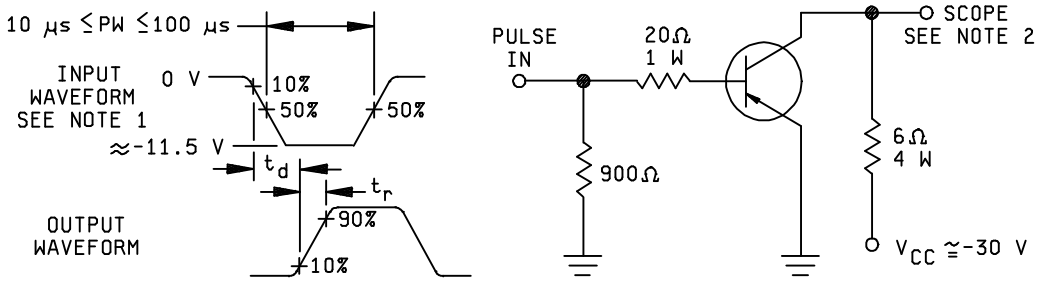
3/ The electrical measurements for appendix E, table VII of MIL-PRF-19500 are as follows:

a. Subgroup 2, See table II herein, steps 1, 2, 3, and 4 (JANS); 1 and 4 (JANTX and JANTXV).
b. Subgroup 3, See table II herein, steps 1, 2, 3, and 4 (JANS); 1 and 4 (JANTX and JANTXV).
c. Subgroup 6, See table II herein, steps 1, 2, 3, 4, 5, 6, and 7 (JANS); 1, 4, 5, and 6 (JANTX and JANTXV).

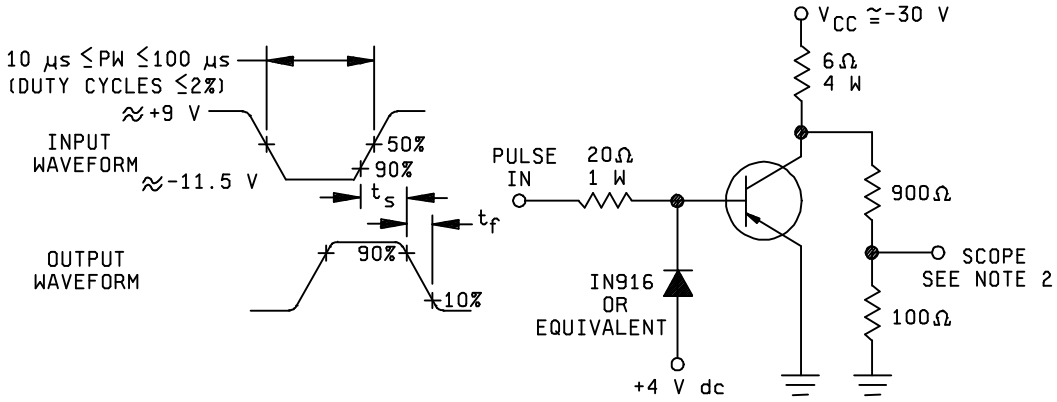
4/ Devices which exceed the group A limits for this test shall not be acceptable.



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TEST CIRCUIT A




TEST CIRCUIT B

NOTES:

1. The input waveform is supplied by a pulse generator with the following characteristics:
   $t_r \leq 2.0$ ns, $t_f \leq 1$ µs, 10 µs ≤ PW ≤ 100 µs, $Z_{OUT} = 50\Omega$, duty cycle ≤ 2 percent.
2. Output waveforms are monitored on an oscilloscope with the following characteristics:
   $t_r \leq 5$ ns, $Z_{in} \geq 100$ kW, $C_{in} \leq 12$ pF.
3. Test circuit A for $t_d$ and $t_r$; test circuit B for $t_s$ and $t_f$.

FIGURE 2. Pulse response test circuits.



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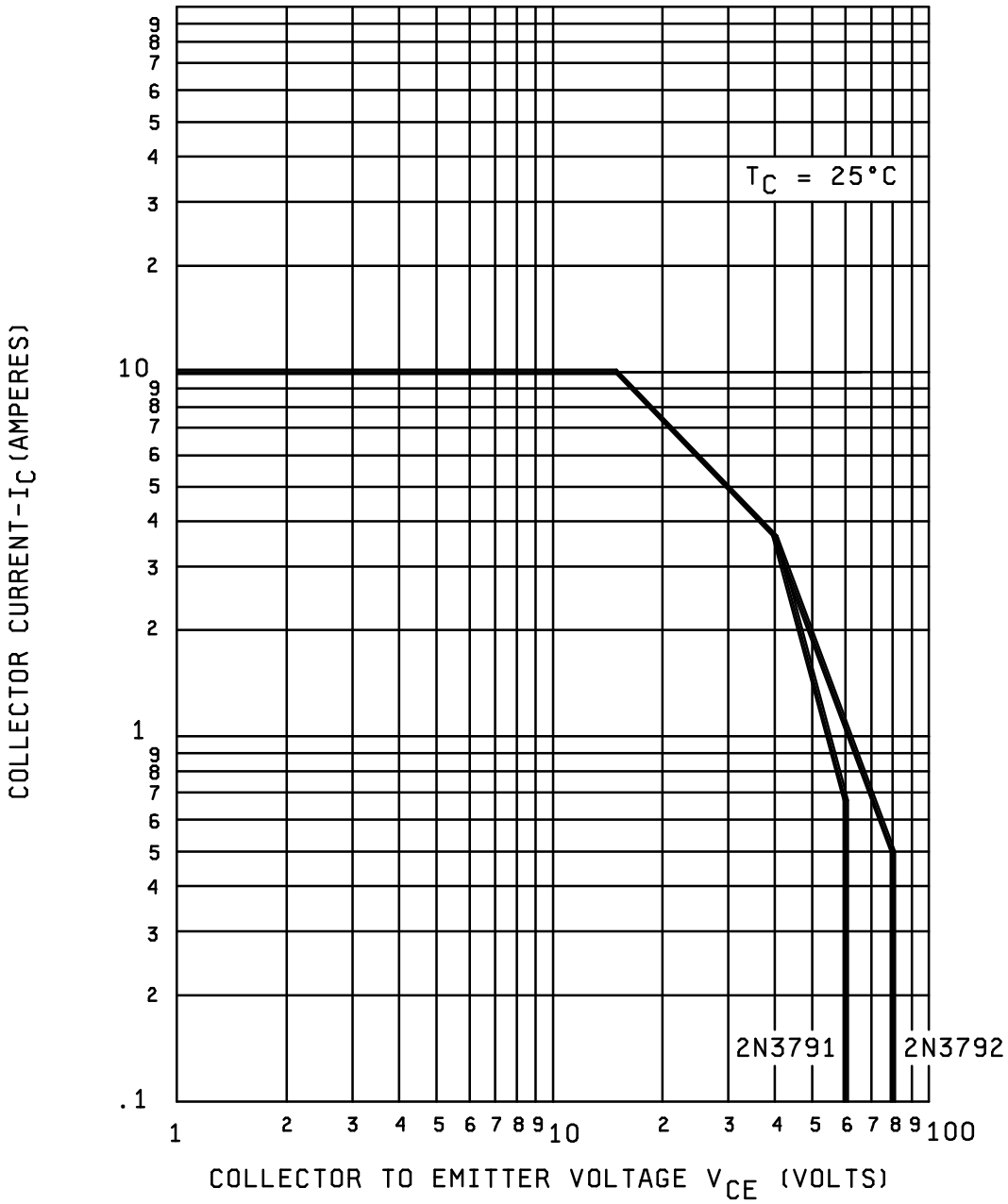



FIGURE 3. Maximum safe operating graph (dc).



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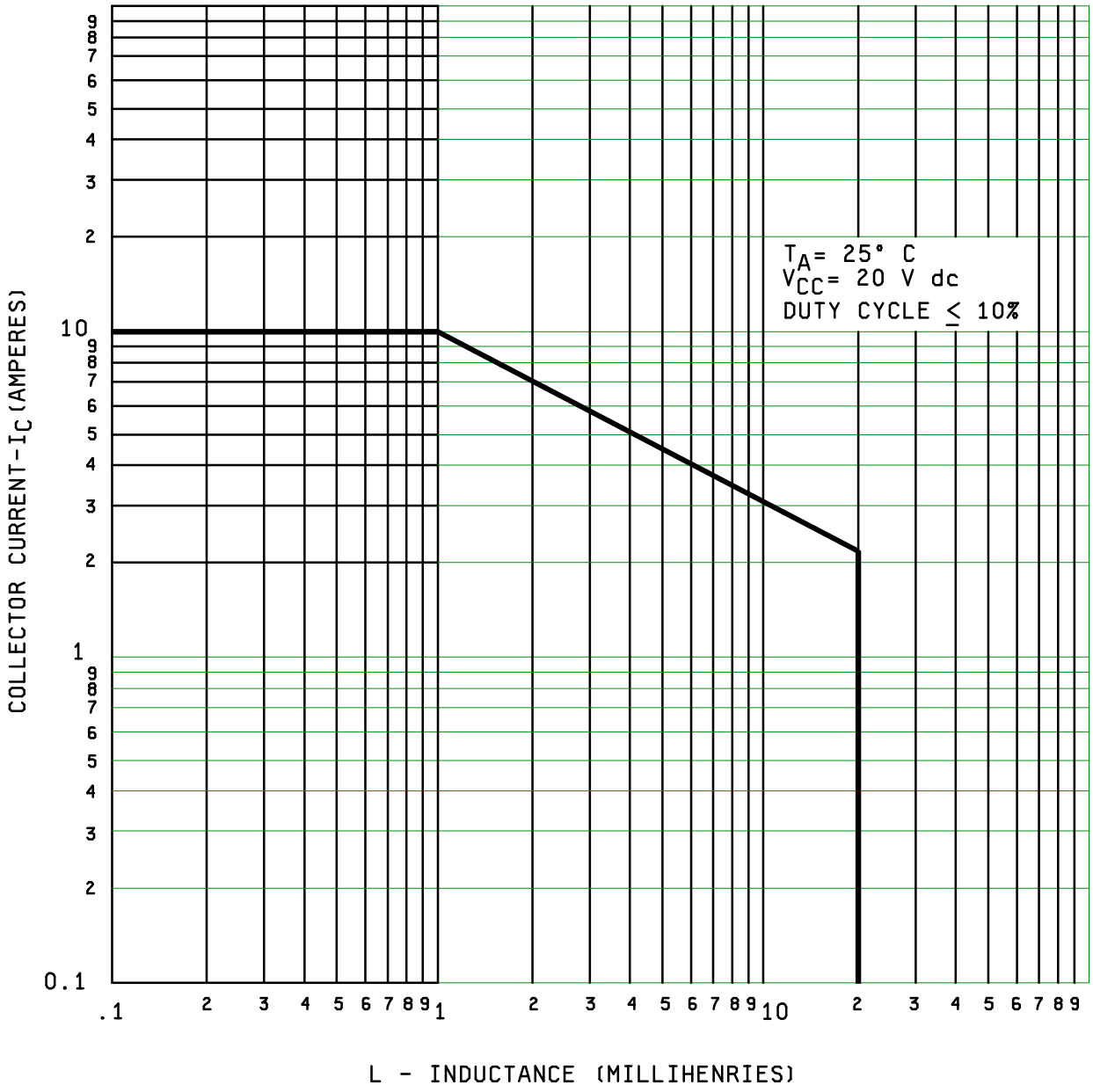



FIGURE 4. Safe operating area for switching between saturation and cutoff (unclamped inductive load).



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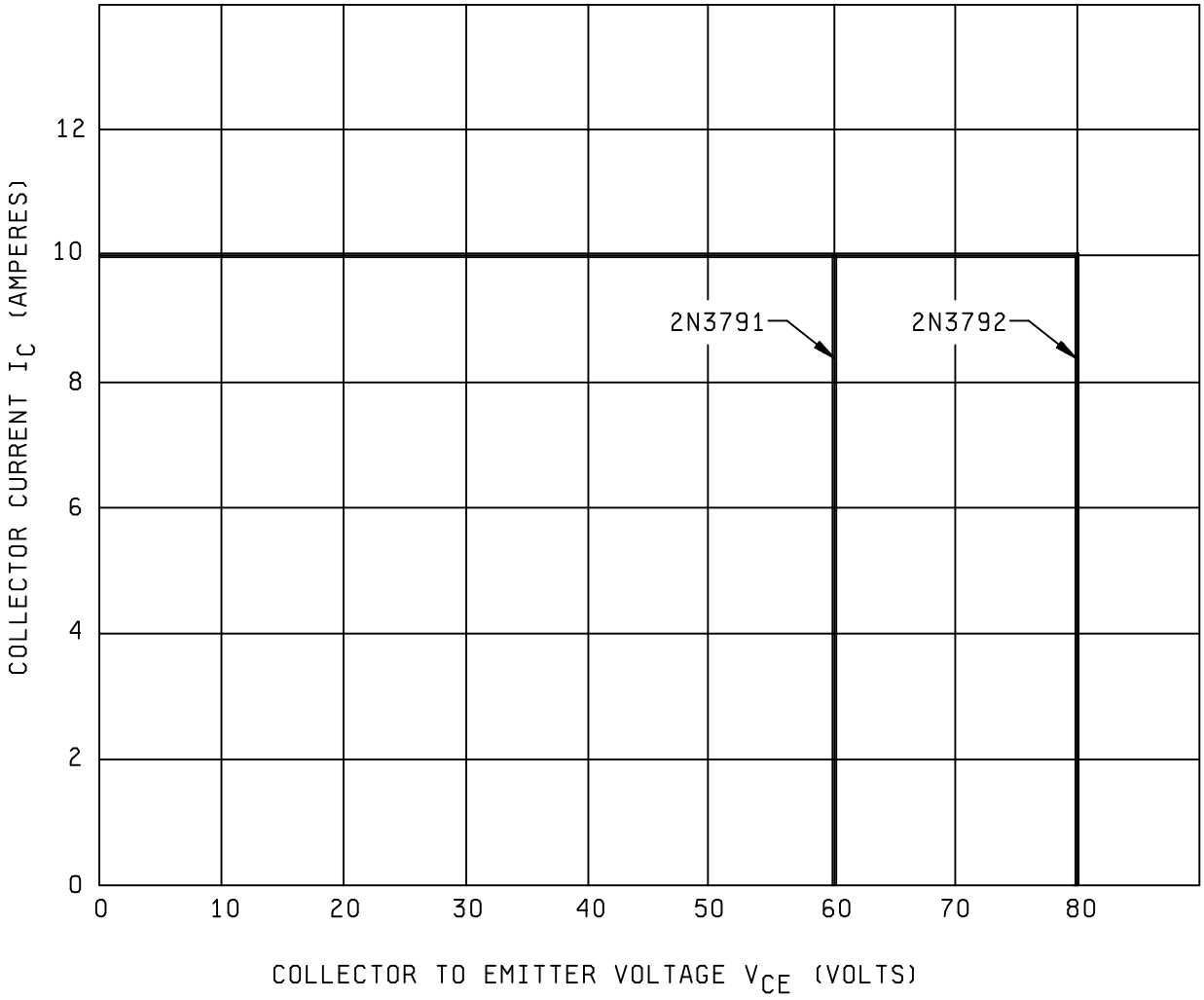
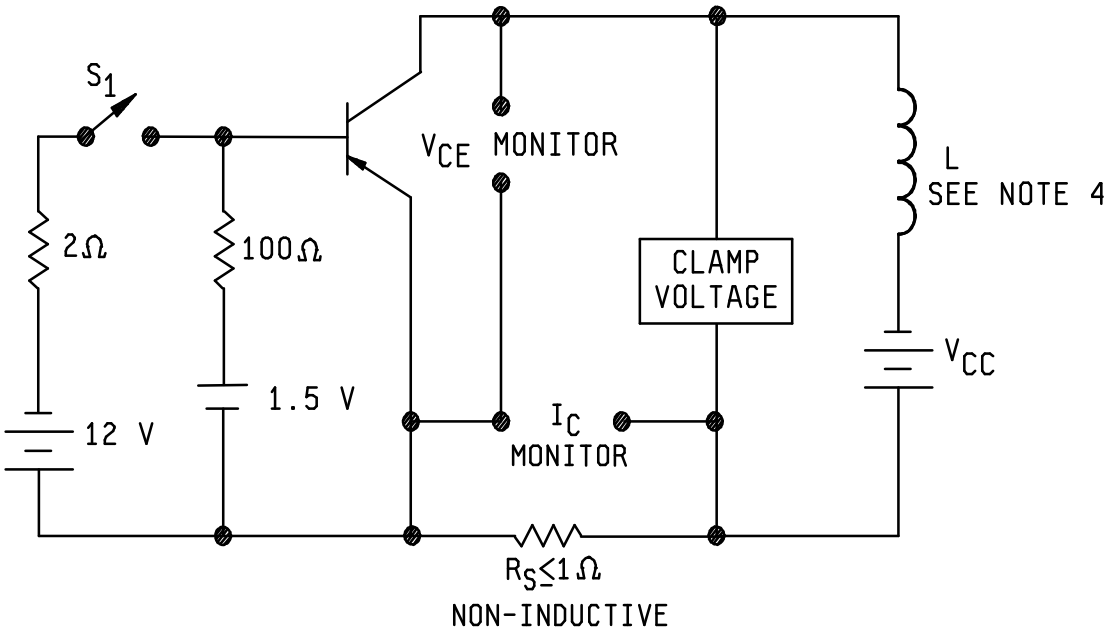



FIGURE 5. Safe operating area for switching between saturation and cutoff (clamped inductive load).



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Procedure:

1. With switch $S_1$ closed, set the specified test conditions.
2. Open $S_1$. Device fails if clamp voltage not reached.
3. Perform specified end point tests.
4. L = 4 mH, .05W, 20 A
   Q | 100 at 1 kHz
   (Stanford Miller CK-20 or equivalent)

FIGURE 6. Clamped inductive sweep test circuit.



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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by Do personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Application guidance. Transistor types 2N3789 and 2N3790 were deleted by MIL-PRF-19500/379A(ER). The following show the replacement types:

| Deleted transistors | Replaced by |
|---|---|
| 2N3789 | 2N3791 |
| 2N3790 | 2N3792 |

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List PL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from , Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue because of the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Air Force - 13, 19, 85, 99
Army - AR, AV, MI, SM
Navy - AS, CG, MC

Preparing activity:
DLA - CC

(Project 5961-1828)



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**STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL**

**INSTRUCTIONS**

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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| **I RECOMMEND A CHANGE:** | **1. DOCUMENT NUMBER** MIL-PRF-19500/379E | **2. DOCUMENT DATE (YYMMDD)** 970925 |
|---|---|---|

**3. DOCUMENT TITLE**
SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, HIGH-POWER, TYPES 2N3791, 2N3792, JANTX, JANTXV, AND JANS

**4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)**

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME (Last, First, Middle initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED (YYMMDD) |

**8. PREPARING ACTIVITY**

| a. NAME Alan Barone | b. TELEPHONE (Include Area Code)<br>(1) Commercial (614)692-0510<br>(2) AUTOVON 850-0510 |
|---|---|
| c. ADDRESS (Include Zip Code) , Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000 | **IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:**<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |



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